

MANNONDÉ DESCOS PRUDENCIO GBAGUIDI

Mathematician · Biostatistician · Data Scientist

Born 17 May 2000 in Savalou, Benin | 6 Melrose Road, Muizenberg, 7945, Cape Town, South Africa

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PROFILE

Young researcher in mathematics, currently completing a Master of Science in Mathematical Sciences at the African Institute for Mathematical Sciences (AIMS South Africa, degree awarded by Stellenbosch University). Holder of two Master's degrees from the University of Abomey-Calavi (Biostatistics; Pure and Applied Mathematics) and a Bachelor's degree in Fundamental Mathematics. Strong experience in mathematical and epidemiological modeling, numerical analysis of PDEs, machine learning, and scientific programming (Python, R, Matlab, C/C++). Actively involved in training, mentorship, and scientific outreach across Africa (AIMS, Africa Scientifique, WAFIRA, Google DeepMind AI Research Foundations).

ACADEMIC EDUCATION

Master of Science in Mathematical Sciences 09/2025 – 07/2026 · Cape Town, South Africa

African Institute for Mathematical Sciences (AIMS South Africa) degree awarded by Stellenbosch University · EQF Level 7

- Msc supervisor: Prof. Dr. Duván Cardona Sanchez, King Fahd University of Petroleum and Minerals (KFUPM), Saudi Arabia
- Specialization: Mathematical Sciences. Program successfully completed in July 2026; official diploma to be conferred by Stellenbosch University.
- Research project: Pseudo-differential Operators and Microlocal Analysis.
- Academic Excellence Award; Class Representative, Master's Cohort.

Master's Degree in Bio-Statistics 09/2023 – 2025 · Abomey-Calavi, Benin

Faculty of Agronomic Sciences (FSA), Doctoral School of Agronomic and Water Sciences, University of Abomey-Calavi (UAC) · EQF Level 7

- Msc supervisor: Professor Dr. Têlé Jonas DOUMATE, University of Abomey-Calavi (UAC), Benin
- Research project: Mathematical modeling of epidemiological disease (Cholera)
- Master's thesis currently under defense.

Master's in Pure and Applied Mathematics 02/2023 – 08/2025 · Abomey-Calavi, Benin

Faculty of Sciences and Technology (FAST), University of Abomey-Calavi (UAC) · EQF Level 7

- Msc supervisor: Professor Dr. Têlé Jonas DOUMATE, University of Abomey-Calavi (UAC), Benin
- Research project: Non-local multi-strain epidemic model
- Master's thesis currently under defense.

Bachelor's Degree in Fundamental Mathematics 11/2017 – 11/2021 · Abomey-Calavi, Benin

Faculty of Sciences and Technology (FAST), UAC Major: Mathematics and Applications, Specialty: Fundamental Mathematics · EQF Level 6

PUBLICATIONS

- Gbaguidi, M. Directional ellipticity and parametrix for fractional Bessel potential with drift. To appear in *Trends in Mathematics*, Springer-Birkhäuser, 2027.
- Gbaguidi, M.; Cardona Sanchez, D. Regularity of solutions for fractional diffusion operators with drift on L^p , Besov and Sobolev spaces.

SCHOLARSHIPS & AWARDS

African Institute for Mathematical Sciences (AIMS South Africa) 2026

- Academic Excellence Award

Mastercard Foundation Scholarship 2025 – 2026

- Prestigious scholarship for study and research at the African Institute for Mathematical Sciences (AIMS).

DAAD Scholarship (Deutscher Akademischer Austauschdienst) 2023 – 2025

- Academic excellence scholarship for research in biostatistics.

World Bank Scholarship 2023 – 2025

- Financial support based on academic excellence for research in pure and applied mathematics.

PROFESSIONAL EXPERIENCE

Research Internship 2026 – ongoing

ICMAM Analysis and PDE Group for Developing Countries

- Research internship within the Analysis and PDE Group for developing countries.

- Supervisor: Prof. Dr. Duván Cardona Sanchez, King Fahd University of Petroleum and Minerals (KFUPM), Saudi Arabia.

Intern in Applied Mathematical Modeling and Data Analysis 09/2024 – 10/2024 · Ekiti State, Nigeria

International Centre for Applied Mathematical Modelling and Data Analytics (ICAMMDA), Federal University Oye-Ekiti

- Practical experience in R programming, malaria modeling, data analysis, and project management.
- Skills developed: statistical modeling, scientific presentation, research networking.

Mathematics Teacher 2021 – 2022 · Zinvié, Benin

Collège Sainte Thérèse de Zinvié

- Supervised "Première D" students in mathematics: geometry, trigonometry, and analysis.

Intern in Mathematical Analysis 08/2022 – 10/2022 · Abomey-Calavi, Benin

Department of Mathematics, University of Abomey-Calavi (UAC)

- Solved analysis problems related to operators on Hilbert spaces.

RESEARCH PROJECTS

Pseudo-differential Operators and Microlocal Analysis 2025 – ongoing

- Study of pseudo-differential operators within the framework of microlocal analysis: symbolic calculus, regularity properties in Sobolev spaces, propagation of singularities of PDE solutions via wave front set theory. Implications for spectral analysis and the design of efficient numerical solvers.
- Supervisor: Prof. Dr. Duván Cardona Sanchez, King Fahd University of Petroleum and Minerals (KFUPM), Saudi Arabia.

Structural Robustness Analysis of an SEIR-B Model for Cholera 07/2023 – 09/2025

- Development and analysis of a compartmental SEIR-B model for cholera dynamics in West Africa. Computation of the controlled reproduction number R_c with sensitivity analysis.
- Supervisor: Prof Dr. Têlé Jonas DOUMATE, University of Abomey-Calavi (UAC), Benin

Multi-strain Epidemic Model with Non-local Dispersion 11/2024 – 08/2025

- Rigorous mathematical analysis of an SIS model with non-local dispersion, governed by integro-differential equations. Numerical exploration of a two-strain model revealing spatial coexistence phenomena through niche segregation.
- Supervisor: Professor Dr. Têlé Jonas DOUMATE, University of Abomey-Calavi (UAC), Benin

Nonparametric Kruskal-Wallis Test in MATLAB, R, and Python 05/2024 – 06/2024

- Comparison of the implementation of the nonparametric Kruskal-Wallis test across three programming languages.
- Supervisor: Full Professor Romain Glèlè Kakai

TECHNICAL SKILLS

Programming & Tools

- Python (PyTorch, TensorFlow, Pandas, Scikit-learn)
- R, Matlab, C/C++
- LaTeX (advanced)
- Microsoft Office Suite, CSPro, Access
- SageMath, Scilab

Mathematics & Data Science

- Stochastic calculus, functional analysis
- Operator theory, microlocal analysis
- Numerical analysis of PDEs
- Epidemiological and biostatistical modeling

CERTIFICATIONS & ADDITIONAL TRAINING

AI Research Foundations for Higher Education Programme Pilot Workshop 2026 · Muizenberg, South Africa

AIMS South Africa, in partnership with Google DeepMind, Google.org, FATE Foundation, UCL Computer Science & UCL Centre for AI

- "Train the Trainer" program covering the technical content of the Google DeepMind AI Research Foundations course, innovative pedagogical practices, and AI teaching approaches.

Elements of AI Online Training Certificate

Reaktor & University of Helsinki (co-signed attestation by a Sorbonne Université representative)

- Comprehensive online training on the fundamentals of artificial intelligence.

Certificate of Training Completion

The Tony Elumelu Foundation

- Entrepreneurship and skills development training program.

Certificate 1st Prize (R1,500) 13–17 April /2026

AIMS South Africa Integrative Think Tank

WORKSHOPS, SEMINARS, CONFERENCES & SUMMER SCHOOLS

- Participant Mathematics in Industry Study Group (MISG), Johannesburg, South Africa (Jan. 2026). Solved industrial problems through mathematical modeling (corrosion inhibition, deep-sea mining extraction). Selected from the AIMS cohort for the preparatory "Graduate Modelling Camp."
- ICAMMDA School on Machine Learning Modelling, Federal University Oye-Ekiti (FUOYE), Nigeria (3–21 August 2026), with Prof. Christopher Thron (Texas A&M University-Central Texas) as guest speaker: foundations of ML/tree-based models, formulating research questions and modeling strategies.
- AI Governance for Policymakers and Regulators three-part webinar, AIMS in partnership with the Smart Africa Digital Academy (SADA), starting 31 August 2026.
- Selected for the first professional training cohort of the new AIMS Senegal campus in Dakar (2026).
- International Conference on Mathematical Sciences and Optimization (AMSO), Tai Solarin University of Education, Ogun State, Nigeria (26–30 August 2024) Certificate of Participation.
- Africa Scientifique Programme Scholars Program (AIMS South Africa & African Gong), Cape Town (February 2026) Certificate of Participation.
- Certificate of Leadership for planning and leading the post-workshop scientific communication project "Africa Scientifique Programme 2026."
- WAFIRA (Women Advancement Forum for International Research & Academia) Male Peer Champion at the "Advancing Women in Mathematical Sciences" workshop, AIMS South Africa (Oct.–Nov. 2025).
- Volunteer Organization of the WAFIRA 2025 workshop "Advancing African Early-Career Women in Mathematical Sciences" (3 days): logistics and participant support.
- AIMS House of Science Volunteer, Wikipedia Edit-a-Thon Programme "Amplifying African Knowledge in the Digital" (July 2026).
- AIMS House of Science Volunteer, "Africa Scientifique and Gender Equality (Women in STEM)" workshops.
- Student Assistant "TurtleStitch and African Prints" workshop (AIMS Schools Enrichment Programme).
- EEIA Benin Excellence (Summer School) Deep Learning, YOLOv7, Abomey-Calavi, Benin (June–Sept. 2023). Implementation of computer vision models for object detection; use of TensorFlow/PyTorch and Pandas for data preprocessing.
- Volunteer Mathematics Tutor Mastercard Foundation Community Service (Give-Back), Cape Town (since Sept. 2025).

PROFESSIONAL REFERENCES

- Prof. Dr. Duván Cardona Sanchez MSc Supervisor, King Fahd University of Petroleum and Minerals (KFUPM), Saudi Arabia.
- Prof. Dr. Karin-Therese Howell Executive Director, AIMS South Africa.
- Prof. Dr Tevian Dray: Professor Emeritus, Departments of Mathematics and Physics, Oregon State University.

LANGUAGES

French: Native language (C2) | English: Fluent (C1) | Fon: Native language

INTERESTS

Finance & Tech: financial trading, cryptocurrencies and blockchain, robotics.

Global Affairs: international geopolitics and strategic studies.

Hobbies: video games, anime, and digital culture.